

Towards trustworthy artificial intelligence in musculoskeletal medicine: A narrative review on uncertainty quantification

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Abstract

Introduction: Deep learning (DL) models have achieved remarkable performance in musculoskeletal (MSK) medical imaging research, yet their clinical integration remains hindered by their black-box nature and the absence of reliable confidence measures. Uncertainty quantification (UQ) seeks to bridge this gap by providing each DL prediction with a calibrated estimate of uncertainty, thereby fostering clinician trust and safer deployment.

Methods: We conducted a targeted narrative review, performing expert-driven searches in PubMed, Scopus, and arXiv and mining references from relevant publications in MSK imaging utilizing UQ, and a thematic synthesis was used to derive a cohesive taxonomy of UQ methodologies.

Results: UQ approaches encompass multi-pass methods (e.g., test-time augmentation, Monte Carlo dropout, and model ensembling) that infer uncertainty from variability across repeated inferences; single-pass methods (e.g., conformal prediction, and evidential deep learning) that augment each individual prediction with uncertainty metrics; and other techniques that leverage auxiliary information, such as inter-rater variability, hidden-layer activations, or generative reconstruction errors, to estimate confidence. Applications in MSK imaging, include highlighting uncertain areas in cartilage segmentation and identifying uncertain predictions in joint implant design detections; downstream applications include enhanced clinical utility and more efficient data annotation pipelines.

Conclusion: Embedding UQ into DL workflows is essential for translating high-performance models into clinical practice. Future research should prioritize robust out-of-distribution handling, computational efficiency, and standardized evaluation metrics to accelerate the adoption of trustworthy AI in MSK medicine.

Level of Evidence: Not applicable.

Abbreviations: AI, artificial intelligence; CP, conformal prediction; CT, computed tomography; DL, deep learning; ECE, expected calibration error; HLA, hidden layer analysis; K&L, Kellgren and Lawrence (osteoarthritis grading); MC, Monte-Carlo; MRI, magnetic resonance imaging; MSK, musculoskeletal; OOD, out-of-distribution; THA, total hip arthroplasty; TRIPOD-AI, Transparent Reporting of a multivariable prediction model for Individual Prognosis Or Diagnosis; TTA, test-time augmentation; UQ, uncertainty quantification; XAI, explainable artificial intelligence.

Amir M. Vahdani and Moein Shariatnia contributed equally to this study and are co-first authors of this work.

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KEYWORDS

artificial intelligence, deep learning, musculoskeletal, trustworthy AI, uncertainty quantification

INTRODUCTION

The past two decades have seen an ever-increasing adoption of automated modelling in healthcare, and in particular, using deep learning (DL; Appendix A) models for medical image processing. This stems from the concurrence of several trends, including the growing availability of medical imaging data, better hardware, and amplified investment in research and development [18].

As a discipline deeply intertwined with utilizing various modalities of imaging, musculoskeletal (MSK) medicine stands to benefit significantly from these advancements. Using DL has already been researched extensively in tasks such as fracture detection [70], septic arthritis assessment [64], joint implant identification [92] and loosening detection [55], lower limb alignment analysis [41, 104], syn-desmotic instability detection [12], spine segmentation [66, 94] and exploratory data analysis refinement [78], among a plethora of others [4, 13].

As with other technologies, trustworthiness is pivotal if DL models are to be integrated into routine clinical care. This is especially challenging given the black-box nature of DL models and the difficulty of understanding how they may have arrived at a particular prediction and how reliable it might be [10–14]. These issues are even more pronounced in clinical scenarios in which high-stakes decision-making has to be coupled with transparency and safety; moreover, in medical practice, these DL models are, somewhat mistakenly, compared to classic statistical models, which are already well-established in medical literature and easier to understand for clinicians [26, 85].

Considerable effort has been made to make DL models trustworthy in recent years [3]; these efforts also include the use of DL models in medical applications, which necessitate reliability and involve several stakeholder parties such as clinicians, patients, and health service providers. Trustworthy artificial intelligence (AI) encompasses several principal aspects [61], including:

- **Explainability:** Also termed ‘explainable AI’ (XAI), refers to specific techniques/algorithms implemented to make the decision-making process behind a DL model’s prediction understandable to the human end-user. An example of explainability for a DL model is the integrated gradients technique [67], which provides visual heatmaps of the areas in an image affecting the model’s prediction the most (equivalent to where the model is paying the most

‘attention’); we will see an example of this technique applied to MSK imaging in Section ‘Total hip arthroplasty (THA) automatic implant detection’ [92].

- **Interpretability:** While in many cases used interchangeably with explainability, we prefer to use interpretability as a term referring to the intrinsic degree of simplicity in understanding how and why a model generates its predictions [61]. A well-known example of an interpretable model from medicine is the CHADS2-VASc score [25]: based on the model, the risk for stroke is calculated by simply summing up seven different indicators. In contrast, DL models consist of millions, if not billions, of parameters and perform an incomprehensible amount of arithmetic operations for each prediction. Arguably, this lack of interpretability is a major barrier to the clinical adoption of DL models; in essence, frameworks such as explainable AI and uncertainty quantification (UQ; explained below) are efforts to alleviate this shortcoming of DL models.
- **Transparency:** Refers to efforts in disclosing as much information as possible on DL models deployed as software services, such as the data on which these models were trained. Emphasis on transparency in research is well-established in medicine, with several widely recognized guidelines that research teams are encouraged to adhere to depending on the type of their study, including the TRIPOD-AI guideline [20] for using AI models in clinical research.
- **UQ:** Providing reliable estimates of *predictive confidence*, that is, how confident a DL model is in its predictions. This is parallel to how clinicians can also communicate their levels of confidence in a diagnosis (or any other form of hypothesis), which lets others know the reliability of said diagnosis. Put simply, UQ methods aim to enhance DL models with the same capability (closely related to confidence intervals in statistics [27]). In the following sections, we provide an in-depth exposition of uncertainty from a medical perspective and its quantification. Recent research has also focused on integrating the UQ and XAI frameworks [71, 98], considering their pivotal role in enabling informed decision-making using DL models.

This article explores how UQ has been applied in MSK imaging, emphasizing its clinical significance (Figure 1). Through illustrative examples, we discuss the benefits and challenges of implementing UQ, advocating for its essential role, not as a technical consideration, but a clinical necessity. We have attempted our best to provide intuitive explanations of

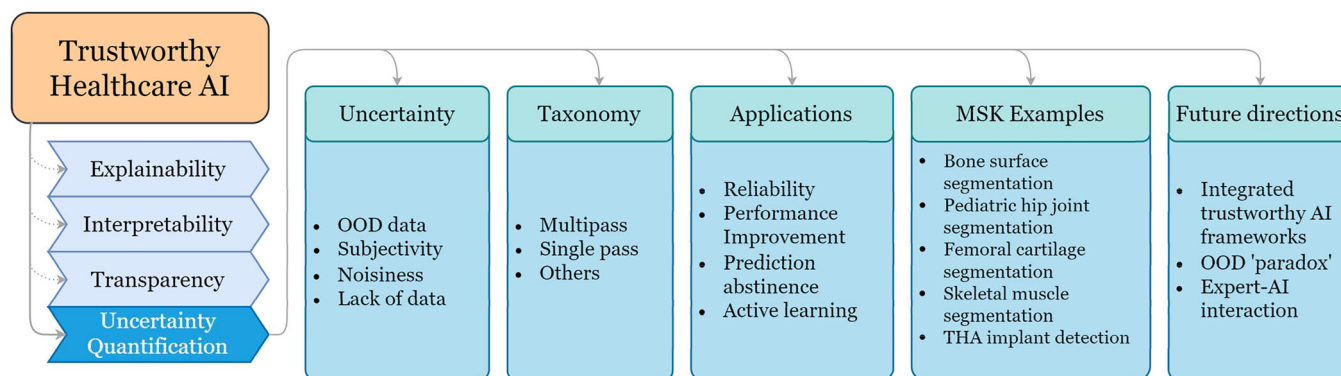


FIGURE 1 Overall structure of the paper and how it fits into the larger trustworthy AI paradigm. AI, artificial intelligence; OOD, out-of-distribution.

UQ methods for a non-technical audience, leveraging analogies and examples from clinical practice; despite this, readers uninterested in technical details could skip Section 'Approaches to uncertainty quantification', while still gaining an understanding of the benefits of incorporating UQ in clinical DL model development. The rest of the paper is organized as follows:

- Section 'Materials and methods' explains our approach to literature review and how relevant studies were included.
- Section 'Uncertainty from a medical perspective' explains uncertainty and its sources from a medical viewpoint.
- Section 'Approaches to uncertainty quantification' outlines our proposed taxonomy for UQ methods and describes each of these methods.
- Section 'Applications of uncertainty quantification in musculoskeletal imaging' provides prominent examples from MSK medicine research utilizing UQ methods.
- Section 'Discussion' provides our conclusions and remarks on the landscape of UQ in medical research.
- Appendix A provides definitions for relevant technical terminology.

MATERIALS AND METHODS

This review is narrative in nature and was developed based on the collective expertise of the author team. Rather than employing a comprehensive systematic search protocol, relevant literature was identified through a selective search of major electronic databases (e.g., PubMed, Scopus and arXiv) and by reviewing references from recent publications in the field of UQ in MSK imaging. Articles were included based on their perceived relevance to the topic and their contribution to the discussion on UQ in DL models for MSK applications. The inclusion criteria

were primarily driven by expert judgement and the integration of insights from other review articles, aiming to provide a focused and clinically relevant overview.

UNCERTAINTY FROM A MEDICAL PERSPECTIVE

Clinicians constantly grapple with uncertainty due to patient variability, imaging artifacts and the inherent limitations of diagnostic imaging, among many other factors. They rely on expressing and understanding this uncertainty to make informed decisions, balance risks, and communicate effectively with patients and colleagues.

Consider the scenario of a radiologist interpreting a knee X-ray image of a potential fracture. If the image is clear and the fracture is evident, the radiologist can confidently make a diagnosis. However, in cases where the image is ambiguous—perhaps due to poor image quality, overlapping structures, or subtle fractures—the radiologist must assess the degree of uncertainty. They might recommend additional imaging, refer to another specialist, or monitor the patient over time. This nuanced decision-making process hinges on the ability to evaluate and communicate uncertainty.

Similarly, for AI models to be truly useful in clinical practice, they must provide not only predictions but also an indication of confidence in those predictions. A model that predicts a fracture with 95% confidence conveys a different message than one that predicts it with 60% confidence. The former might be sufficient to proceed with treatment, while the latter may prompt further investigation. Therefore, integrating UQ into AI models aligns them more closely with clinical reasoning and decision-making processes.

Typically, a DL model outputs a single value for each subject between 0 and 1, with values closer to 1

indicative of the 'positive' class, such as 'fracture positive' in the scenario mentioned above. Since these values are between 0 and 1, they can be interpreted as *probabilities*, which measure the likelihood of something happening, or in this case, being the actual diagnosis of the subject; such outputs are, in most cases, *miscalibrated* in their unmodified form [36]: in this scenario, if the model predicts a 20% probability of fracture for a set of cases, approximately 20% of those cases should actually have fractures. This alignment between predicted probabilities and observed empirical frequencies, known as *calibration*, is essential for reliable clinical decision-making. Nevertheless, a model can be perfectly calibrated, but also useless if it is non-discriminative; for example, if it just outputs the prevalence of fractures as the predicted probability, without taking into account the features of a subject. This trivial solution is also known as *mode collapse*.

In a real-world clinical scenario, several factors can cause these predictions to be *uncertain* (ideally, this would be indicated by predicted probabilities that are far from 0% or 100%). Possible sources of uncertainty in clinical reasoning (e.g., diagnosis) include:

- A new test subject might come from a different *distribution* compared to the data set the model was trained on (also known as out-of-distribution, or OOD), due to:
 - Demographic factors: for instance, a different age group the model has not seen in its training data (think of a medical student only familiar with adult radiographs, seeing paediatric radiographs for the first time), or a different race.
 - Clinical factors: chronic conditions, such as osteoporosis in this case and anatomic variations.
 - Imaging-related factors: These include differences in hardware and software used to procure the image, which might affect characteristics such as brightness, contrast and noise, as well as variations in the imaging procedure, such as posture and angle. In fact, it is possible someone might unknowingly use our hypothetical model on a *lateral* knee radiograph.
 - A new test subject might belong to a different *class*, which is another example of OOD data. Based on our example scenario, this could be a possible diagnosis outside the space of {fracture positive, fracture negative}, such as bone tumors. Again, think of a physician encountering an entirely new medical condition for the first time in a radiograph.
- Human *subjectivity*: Many imaging datasets require subjective input from human experts to procure their labels. For example, inter-rater variability might lead to differences in the contours of structures with

unclear borders in segmentation data sets when multiple experts label the data set.

- Inherent *noisiness* of the data: Some images might be altogether difficult to diagnose (or in DL terms, *label*) even for an experienced practitioner, or subject to disagreement when assessed by multiple experts; this is common in segmentation tasks, for which the contours of a specific structure of interest (e.g., a tumour) might be unclear.
- Small *sample size*: The learning curve for capturing the characteristics of a specific condition may require a larger number of images in the training data set, due to factors such as heterogeneity (think of encountering a familiar condition in an image, but in a previously unseen anatomic location, or in a different patient demographic). In general, biological systems (e.g., human brains) learn much more efficiently than current DL models, that is, they need much fewer samples to learn [57, 100, 103]. This issue is pronounced in medical research due to difficulties in acquiring medical data, especially in the case of less common conditions, or due to ethics/privacy concerns.

Our aim in highlighting the above is to emphasize the pervasiveness of uncertainty in both DL/statistical modelling and clinical reasoning, as well as the inherent challenges of applying DL models in clinical practice. Despite their accuracy, a significant barrier to clinical adoption of DL models is the absence of reliable UQ. Clinicians routinely evaluate and communicate uncertainty to guide decision-making, particularly in high-stakes scenarios. UQ methods aim to equip DL models with similar capabilities, making them an essential component of the effort to establish DL models as trustworthy tools in clinical settings.

APPROACHES TO UQ

Contemporary medical image processing literature offers a multitude of UQ approaches, which aim to provide estimates of predictive uncertainty/confidence in a variety of tasks such as diagnosis and disease staging. Broadly, these can be divided into three categories:

1. Multi-pass approaches: Several predictions are made for each input subject to quantify the uncertainty for each new input; these predictions are not identical since changes are made to the model or the input during each round, similar to how a radiologist might view an image in different settings (e.g., different brightness settings) or how several radiologists might examine the same image. Since these predictions will vary to some degree, variance among these predictions (using metrics such as standard deviation) would indicate uncertainty.

Also, ample evidence shows that using an average from these predictions leads to superior prediction performance compared to a single point prediction [56, 76, 82].

2. Single-pass approaches: These methods utilize specific modifications to a 'conventional' DL model (which only outputs point predictions) to also predict uncertainty levels alongside their output; these range from changes to the model's architecture, to the training process, and to purely post-hoc procedures. During inference, these models need only one round of prediction, which gives them an advantage in speed and computational cost; nevertheless, training these models can be more difficult and expensive due to their specific implementations.
3. Other/mixed approaches: There are other approaches where a mixture of the above methods is used, or a whole new model with the sole task of measuring uncertainty is developed, which does not necessarily fit in any of the above categories.

Multi-pass approaches

Test-time augmentation (TTA)

At test time, several predictions are made using the same model, but for slightly changed versions of the input image (Figure 2; Several figures from this paper use a sample image from the RSNA Paediatric Bone Age Challenge data set [37], available on RSNA's official website) [9]. If a single, 'final' prediction is needed, these predictions are commonly aggregated via mean or mode calculation, for regression and classification tasks, respectively. These modifications include spatial (e.g., flipping, rotation and zooming) and/or intensity (e.g., brightness, contrast and random noise) changes. This process is akin to how a radiologist might view the same computed tomography (CT) image in different intensity 'windows' and/or zoom levels. TTA is very simple to implement, and does not require any modifications to a previously trained model.

Model ensembling

Using several models instead of one to make predictions is a well-established idea [76, 82]. This approach is analogous to how several physicians might collaborate to establish a diagnosis for a subject, thereby reducing the likelihood of diagnostic errors. In both scenarios, higher degrees of disagreement among predictions (or diagnoses) indicate uncertainty. The models within an ensemble can exhibit varying degrees of difference, ranging from similar models initialized with different random seeds to those with fundamentally distinct

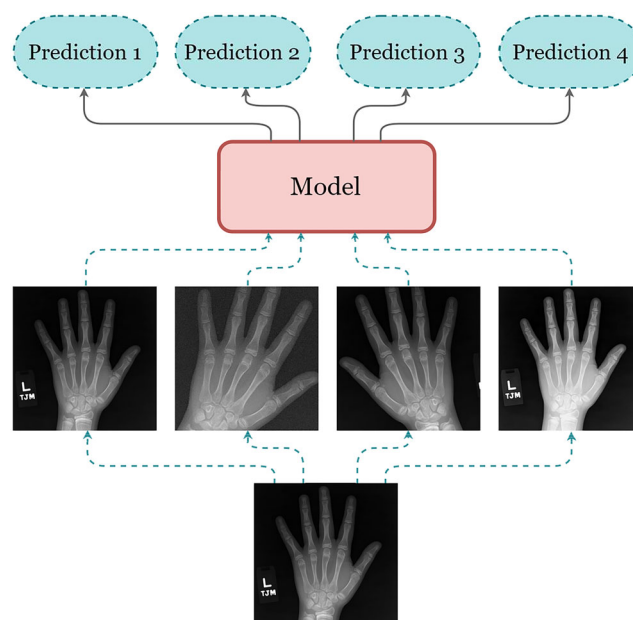


FIGURE 2 Test-time augmentation. Possible modifications include spatial (rotate, flip and zoom) and intensity (brightness and contrast) changes; notice how, regardless of the change, the label for the input (in this case, bone age) remains the same. Most implementations also use an unmodified version of the image for one of the several rounds of inference. Example input adapted from reference [37].

architectures. Higher degrees of difference between the models are linked to improved predictive performance [34, 58]. Nevertheless, implementing model ensembles necessitates the training of several individual models (Figure 3), which may pose limitations in terms of time or computational resources.

Bayesian DL

Bayesian DL tries to apply the principles of Bayesian statistics to DL models. Bayesian statistics is well-known in medical literature and treats probabilities as degrees of belief that can be updated as new information becomes available; think of how there are *prior* beliefs regarding a patient (for instance, based on their demographic), which is updated based on the likelihood of the observed clinical findings to reach a *posterior* conclusion, with the intent of reaching a final diagnosis with sufficient plausibility. Technical details aside, Bayesian DL approaches treat every single parameter in a model as a *distribution* instead of a fixed value, as is common in conventional DL. The prior beliefs regarding these distributions are updated based on the likelihood of the observed training data to acquire posterior beliefs for each parameter. Post-training, each parameter is a statistical distribution (in many cases,

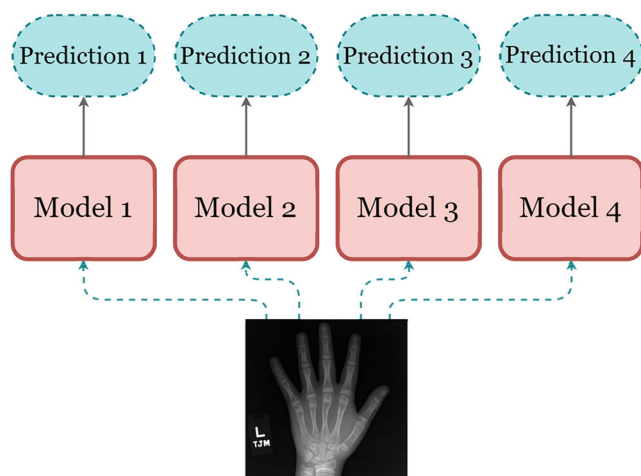


FIGURE 3 Model ensembling. Several separate models are used during inference to obtain multiple predictions for a single input. Input image adapted from reference [37].

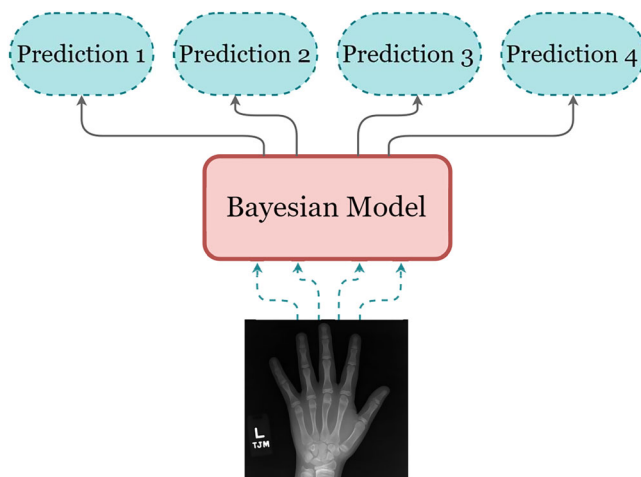


FIGURE 4 Bayesian deep learning. Each parameter is a distribution instead of a fixed value. For each round of inference, these distributions are sampled, which lead to non-identical predictions for a single input. Input image adapted from reference [37].

a normal distribution with mean and standard deviation learned during training), which can be sampled for several rounds of prediction on a single subject; since these samples are random, the predictions are also non-identical (Figure 4), and higher degrees of difference between these predictions indicates uncertainty. Bayesian DL methods are computationally expensive and are less often used in medical image processing compared to most other methods [59]. However, several other UQ approaches (Sections ‘Model ensembling’ and ‘Monte-Carlo (MC) dropout’) are implicit approximations of this method [28, 42, 68, 109] and are more widely used due to more straightforward implementation.

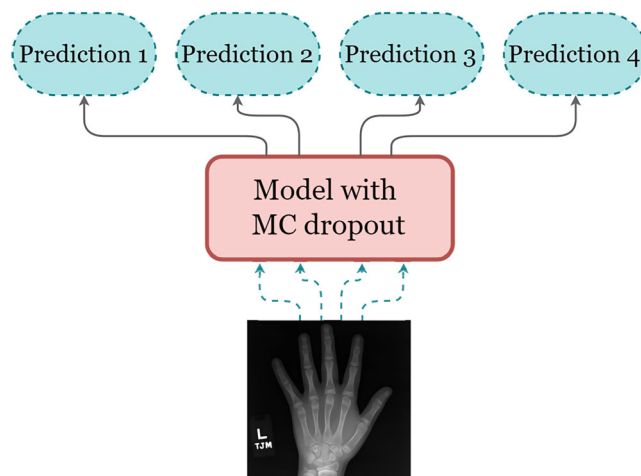


FIGURE 5 Monte-Carlo (MC) dropout. During inference, random outputs from specific hidden layers can be set to 0, with a specific probability (e.g., 0.2). This can lead to non-identical predictions for a single input. Input image adapted from reference [37].

Monte-Carlo (MC) dropout

MC methods refer to the repeated simulation of a random process to estimate a value or distribution that might be problematic to calculate mathematically. MC dropout, also known as test-time dropout, refers to performing dropout during inference as well as training, leading to non-identical predictions each time the model performs inference on a test input (Figure 5). This method is compatible with any previously trained model that has already used dropout in its training and is widely utilized in medical image processing literature [59].

Single-pass approaches

Direct predictive confidence

As stated earlier, DL models trained for classification or segmentation provide outputs in the range of (0, 1) by default, which are often miscalibrated. As such, many studies use these outputs as baseline measures of uncertainty since they require the least effort to obtain. Also available are post-hoc calibration techniques [72, 83, 119]; while these methods entail little computational cost, they do require a separate calibration data set, which might be difficult to collect in medical settings.

Confidence branch

First proposed by DeVries et al. [23], the confidence branch is an additional branch that lets the model ‘ask for hints’ during training, and outputs an additional value between 0 and 1 during inference, with higher values indicative of uncertainty (Figure 6). Since the

model is trained to make a compromise between predicting the correct class and asking for hints, the outputs of the confidence branch can be used as a measure of uncertainty. This method requires changes to the model's architecture and training, but does not add notable computational costs during inference.

Conformal prediction (CP)

CP [5, 99] is a purely post-hoc UQ method that is increasingly being used in medical research [108]. This

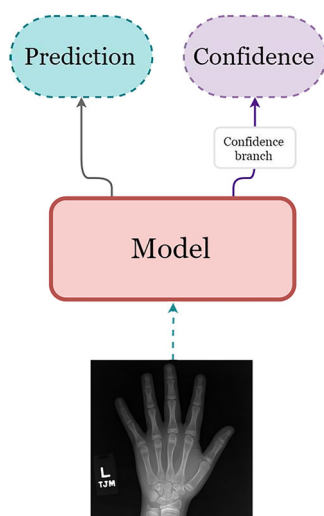


FIGURE 6 Confidence branch. A separate branch from the model is trained to ‘ask for hints’ and also compromise between minimizing the need for hints and prediction accuracy. During inference, this branch outputs a value, c , between 0 and 1; $1 - c$ is used as a measure of confidence. Input image adapted from reference [37].

method requires a separate calibration data set for a previously trained model (in the context of CP, ‘calibration’ refers to a specific procedure distinct from the statistical concept of calibration described in Section ‘Multi-pass approaches’). Using the calibration data set, a distribution of non-conformity scores (indicative of atypicality) can be acquired, which can be used to assess later test data in terms of ‘strangeness’, or non-conformity, compared to the training data set. The fundamental principle of conformal prediction is that, for new test instances that differ significantly from the training data, the associated predictions are expected to be more uncertain. This information is used to refine the predictions of the model: during inference, CP transforms the point predictions of a conventional DL model to a predictive set (for classification) or interval (for regression) using the aforementioned distribution, with a theoretical guarantee of containing the ground truth value with a user-specified level of error (e.g., 5%). As a clinical analogy, these sets could be thought of as a list of plausible differential diagnoses. Notably, CP does not provide an uncertainty estimate per se, but larger predictive sets/intervals imply higher levels of uncertainty. CP can be evaluated using metrics on coverage (how often the prediction sets cover the ground-truth label) and efficiency (how small the sets are). Figure 7 provides examples of the predictions of a model complemented with CP, trained to predict K&L grading [51, 96] for knee radiographs; another example is also available in Section ‘Total hip arthroplasty (THA) automatic implant detection’.

This method offers several attractive properties, such as minimal computational cost and relative theoretical simplicity. However, it is relatively data-hungry since it requires a separate calibration data set on top of the routine training and test data sets.

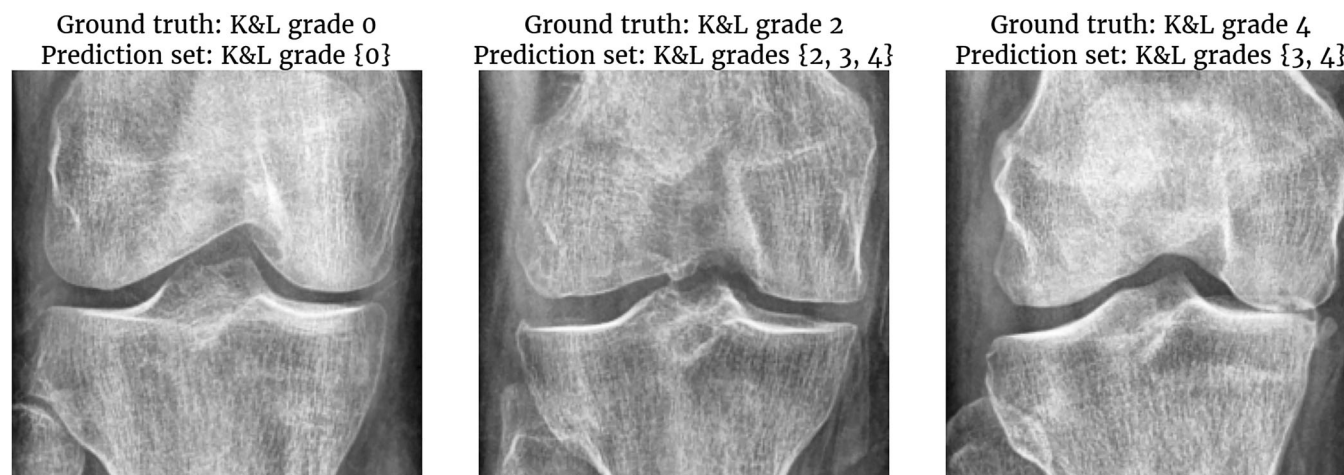


FIGURE 7 Conformal prediction. Example prediction sets from a model trained by the authors on data from reference [19] to predict K&L grading on knee radiographs. Notice the uncertain prediction for the middle image, which shows moderate levels of joint space narrowing. Radiograph images shown are modified using the CLAHE algorithm [95, 123] for better visual clarity. CLAHE, contrast-limited adaptive histogram equalization; K&L, Kellgren and Lawrence.

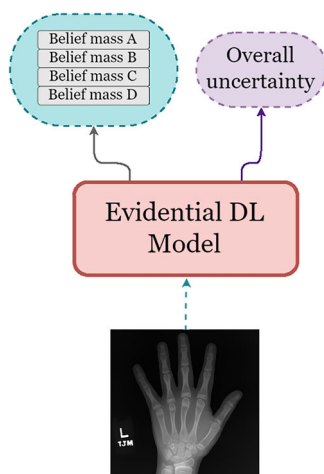


FIGURE 8 Evidential deep learning. The model is able to assign class-wise belief masses, along with an overall uncertainty level for each input. Input image adapted from reference [37].

Evidential DL

The evidential DL framework [97] takes a unique approach to classification problems by explicitly measuring the 'evidence' for each possible class and assigning what are called 'belief masses' to each class. These belief-masses, along with a measure of overall uncertainty, sum to 1 (Figure 8). This differs from conventional DL models (Section 'Uncertainty from a medical perspective'), where predicted probabilities alone sum to 1, without accounting for overall uncertainty. The concept of gathering evidence in support of certain diagnoses is familiar from a clinician's perspective; for example, joint effusion, subchondral bone destruction, and/or joint space narrowing in a radiograph could be considered as 'evidence' supporting the diagnosis of septic arthritis, assigning an increased 'belief mass' to this diagnosis. In both cases (evidential DL and the medical diagnosis process), higher levels of evidence available indicate less uncertainty, and vice versa.

While this method does not require much change to a model's architecture, it does necessitate specific training procedure changes and is thus incompatible with previously trained models.

Learned uncertainty

These methods encompass implementations approaching uncertainty from heuristic perspectives, such as treating inter-rater variability as ground truth for UQ [2, 17, 43, 44, 48, 90, 115] (Figure 9) or including an additional class for the 'uncertain' samples [14] (based on the previously mentioned hypothetical example, going from {'fractured', 'non-fractured'} to {'fractured', 'non-fractured', 'uncertain'}); during training, the model can be trained to predict the correct or 'uncertain' class for each subject.

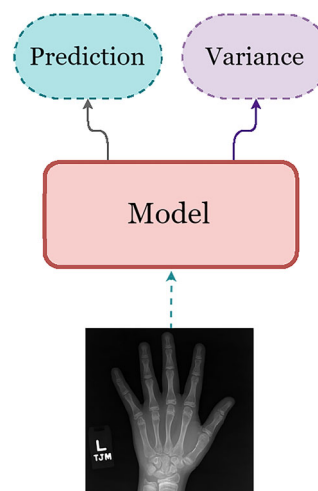


FIGURE 9 Learned uncertainty. This category includes several approaches. In the case of the example shown, the model is trained using multiple ground-truth labels from several experts, and is designed to also learn the heterogeneity of the labels for each input image. During inference, the model outputs a prediction along with a confidence level, based on its predicted level of heterogeneity for the hypothetical labels of unlabelled test data. Input image adapted from [37].

Hidden layer analysis (HLA)

These methods leverage the intermediate, hidden layers of a DL model for UQ, given the unreliability of the raw predicted probabilities for UQ [39, 77] (Figure 10); a variety of algorithms are used to quantify uncertainty or identify unfamiliar subjects based on the outputs of the intermediate layers for each subject.

Other approaches

As mentioned earlier, these include approaches integrating several of the above methods or utilizing separate models that are responsible solely for UQ. For instance, several teams have trained secondary *generative* models to reconstruct the images from the same data set the primary model is trained on (e.g., a data set of T1-Gd brain magnetic resonance imaging [MRI]) (Figure 11); during inference, this secondary model will have worse performance in reconstructing test input images from a different distribution (e.g., a data set of a different MRI sequence), which indicates uncertainty in the predictions of the primary model.

Overall taxonomy

To conclude this section, we provide our overall categorization of the UQ methods discussed in

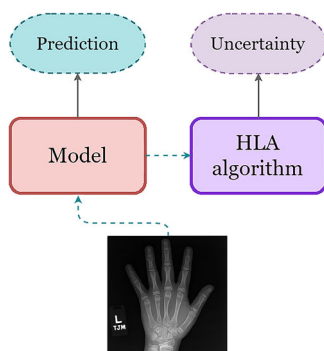


FIGURE 10 Hidden layer analysis (HLA). A secondary algorithm predicts uncertainty levels based on the outputs of the hidden layers of the model; possible options include separate machine learning models or comparing hidden layer outputs for test data against the distributions of the same outputs for training data. Input image adapted from reference [37].

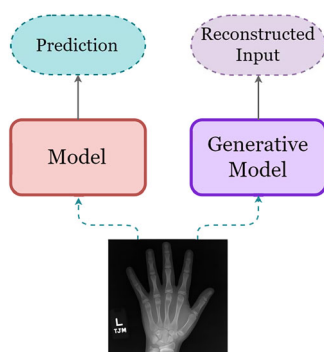


FIGURE 11 Generative approaches. A secondary model is trained on the training data, not to predict the labels, but to reconstruct the input image. For any given test input image, if the reconstructed output differs significantly from the input, this indicates a high level of uncertainty. Input image adapted from [37].

Table 1. This table summarizes several salient characteristics for each method, including the option of post-hoc implementation, how each UQ method affects a DL model overall, and how each of these methods affects the base model in terms of classification/segmentation performance, based on the original works available in the medical DL literature. We have also listed several examples for each method, some of which are outside the scope of MSK imaging.

APPLICATIONS OF UQ IN MSK IMAGING

In this section, we review several papers that showcase the applications of UQ in MSK imaging, highlighting the benefits of UQ for each study's specific use case.

Bone surface segmentation in ultrasound images

Pandey et al. [81] developed U-Net models for segmenting bone surfaces in ultrasound images, a modality known for its operator dependency and susceptibility to artefacts, making uncertainty estimation particularly valuable. They experimented with three of the multi-pass approaches explained in Section 'Approaches to uncertainty quantification', namely MC dropout, TTA, and model ensembling. To assess the generalizability of their model and UQ method on previously unseen data, they evaluated their models on both the full validation data set and an 'adequate only' subset, which excluded ultrasound slices containing no bone surfaces or bone surfaces outside the middle 80% of the image. Essentially, the 'adequate only' subset constitutes 'cleaner' data, which is easier for both segmentation and UQ, compared to the entirety of the data set. They also assessed the UQ methods in terms of estimating predictive confidence (i.e., lower uncertainty is expected to indicate better segmentation performance) and calibration. Figure 12 shows an example from their work, comparing the outputs of an uncalibrated model with a well-calibrated ensemble of 20 U-Net models.

Per their evaluations, the ensemble models achieved the best results in segmentation performance, calibration, and predictive confidence estimation. Nevertheless, confidence estimation was deemed unsatisfactory across all three UQ methods, especially in the full validation data set: correlations between the uncertainty measures and segmentation performance were significant only in the case of the ensembling method when used on the clean 'adequate only' data set. This is consistent with the generally accepted weakness of UQ methods when dealing with OOD data, since at a basic level, UQ methods themselves can be thought of as models/estimators trained on a specific distribution of data [59, 80].

Paediatric hip joint ultrasound segmentation

Kannan et al. [49] compared the performance of a baseline U-Net model against a U-Net model utilizing MC dropout and a modified training procedure designed to focus on uncertain areas in the input image, in order to assess the application of DL in investigating developmental dysplasia of the hip, a condition which necessitates early and accurate diagnosis in order to prevent long-term complications. The researchers found that incorporating uncertainty into the training process improved the

TABLE 1 Overall taxonomy and characteristics of uncertainty quantification methods used in medical deep learning tasks.

Category	Method	Post hoc? ^a	Implementation (before inference)	Implementation (inference)	Effect on prediction metrics	(Other) Examples	Considerations
Multi-pass	Test-time augmentation	Yes	-	Input augmentation	Positive [7, 21, 38, 47, 107, 117, 118]	[7, 21]	Simple to implement
	Ensembling	No	Multiple models trained	Multi-pass inference	Positive [69, 73, 114]	[11, 63, 106]	Simple to implement, high computational cost during training
	Bayesian DL	No	Changes to architecture and training	-	Positive [1, 10, 86, 101]	[84, 102, 113]	High computational cost during training
	Monte-Carlo (MC) dropout	No ^b	Use dropout	Dropout kept active; multi-pass inference	Positive [30, 52]	[7, 11, 15, 54, 93, 105, 106]	Simple to implement
Single-pass	Direct predictive confidence	Yes	Calibration	-	Positive (if calibrated)	[7, 11, 91, 92]	Simple to implement, unreliable in case of no calibration
	Confidence branch	No	Changes to architecture and training	-	Inconclusive [23, 88, 120]	-	-
	Conformal prediction	Yes	Calibration	Prediction sets	-	[22, 24, 65, 79, 92, 121]	Data-hungry
	Evidential DL	No	Changes to architecture and training	Belief mass, overall uncertainty	Positive [29, 35, 45, 46, 53, 122]	[8, 62, 74, 111, 112, 116]	-
	Learned uncertainty	No	Changes to architecture and training	-	Inconclusive [17, 44, 48, 75, 115]	[93]	-
Other/mixed	Hidden layer analysis	Yes	-	-	-	[7, 11, 31, 50, 92, 105, 110]	-
	Generative methods	No	Additional reconstruction model	Reconstruction error	-	[16, 32, 33]	High computational cost due to additional model

Abbreviation: DL, deep learning.
^aPost-hoc methods are compatible with previously developed DL models (possibly by other teams). This is an important consideration given the increasing availability of open-source models in medical research and the significant costs associated with the de novo training of some DL models.
^bWhile MC dropout technically requires a condition (enabling dropout during inference), this condition can be fulfilled with minimal effort in most cases, due to the widespread use of the dropout technique in training most DL models.

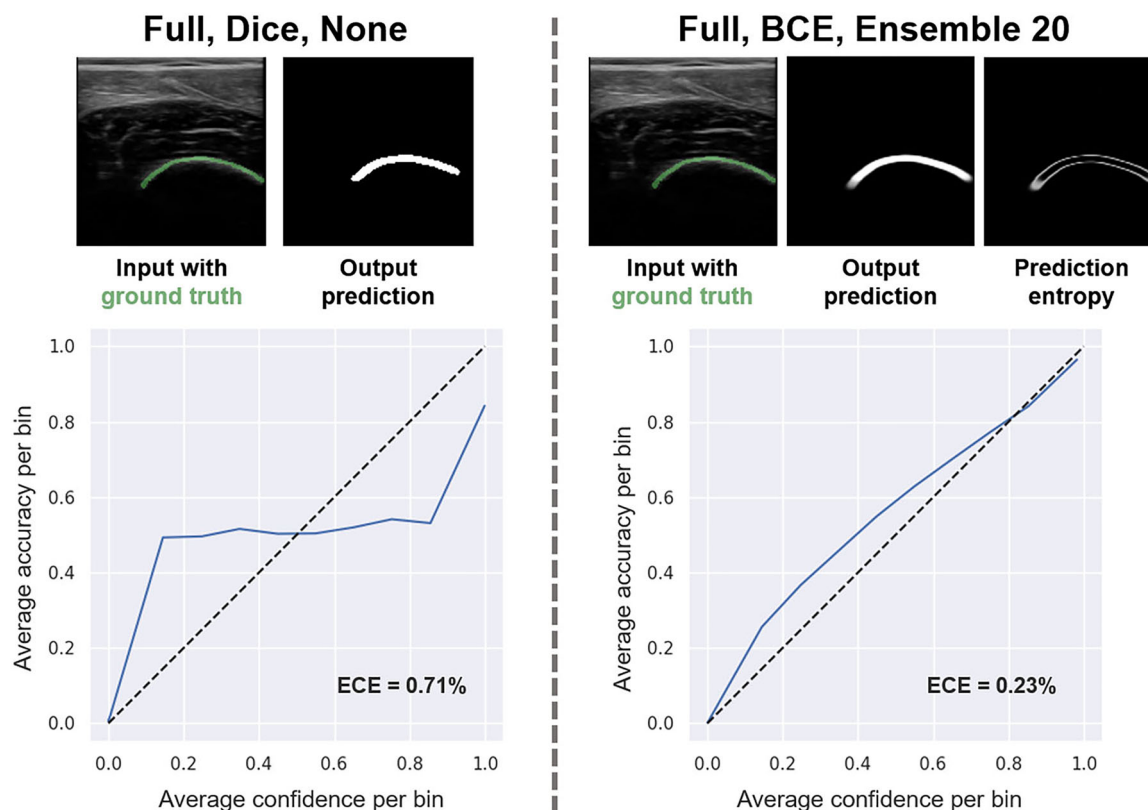


FIGURE 12 Diagram from reference [81] (license number: 5855230400153), comparing predictions from a baseline U-Net model (left) against an ensemble of 20 U-Net models (right). 'Prediction entropy' reflects the level of difference between the 20 predictions in the right plot, which indicates uncertainty (highest around the contours of the prediction). The plots compare the miscalibrated baseline predictions against the ensemble predictions, which have smaller expected calibration error (ECE).

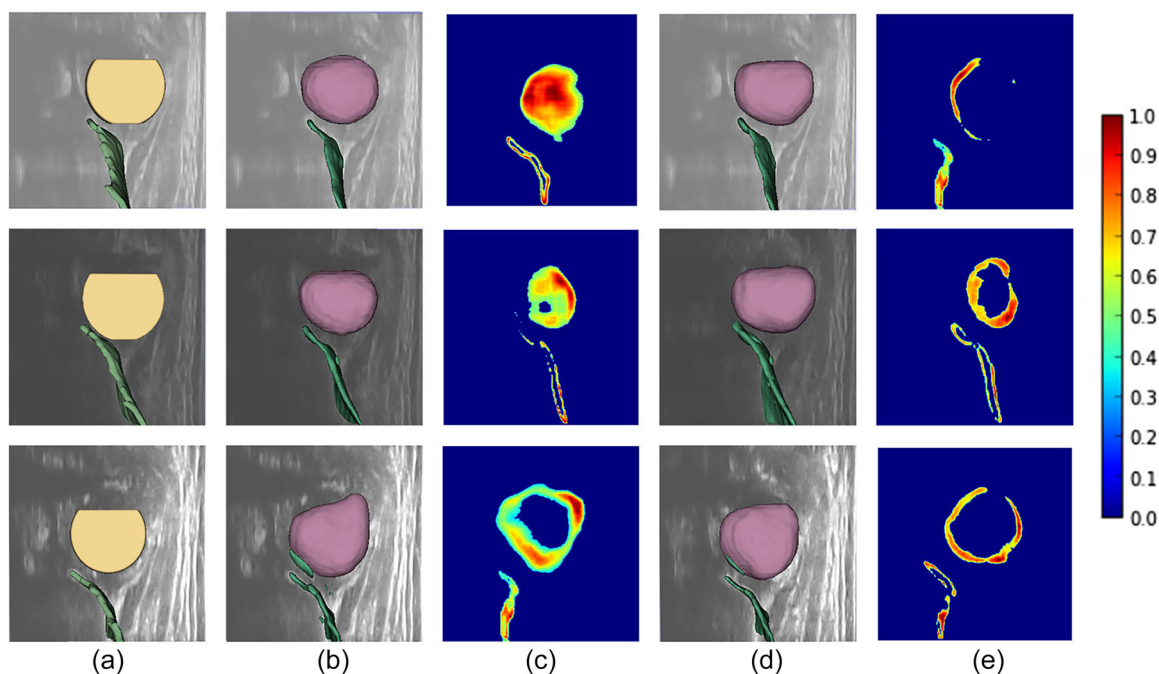


FIGURE 13 Diagram from reference [49] (license number: 5855230576544), showcasing hip joint segmentation. (a) ground truth label. (b) and (d) predictions with different configurations in training procedure. (c) and (e) uncertainty heatmaps for (b) and (d), respectively. Colour bar implies uncertainty levels.

model's performance (Dice score change of +5%). This approach mirrors the clinical practice of giving special attention to ambiguous findings to ensure accurate diagnosis. Figure 13 showcases an example from their work.

Femoral cartilage ultrasound segmentation

Antico et al. [6] utilized MC dropout for UQ in segmenting ultrasound images of the femoral cartilage, intended for guidance in robotic knee arthroscopy. To address the challenges in labelling the femoral cartilage in ultrasound images, they also obtained ground truth labels from aligning MRI images of the same volumes onto the ultrasound images and compared three approaches: baseline model trained on ultrasound image ground truth, baseline model with MC dropout and model with MC dropout trained on ground-truth labels incorporating uncertainty based on both ground truth labels (from ultrasound and MRI images). The MRI-based approach

obtained the best segmentation performance, in which using MC dropout resulted in Dice score increases of 4.3%–12.4% depending on the exact configuration of the MC dropout method and discarding a maximum of 4.8% of the input image's pixels due to uncertainty. The authors also noted how using a multi-pass UQ method (MC dropout in this case) could be detrimental in real-time scenarios since these methods increase inference time proportional to the number of predictions obtained for each input.

The authors also observed that regions with high model uncertainty often coincided with areas where expert annotators faced difficulties. This underscores a key role of UQ: uncovering intrinsic limitations of both the AI model and the imaging modality. In this case, the uncertain regions reflected the inherent limitations of ultrasound imaging in capturing fine anatomical details, as evidenced by discrepancies between ground-truth labels derived from MRI and ultrasound images (Figure 14). Their study demonstrates the potential of integrating UQ with multi-modality data to address these challenges in MSK imaging.

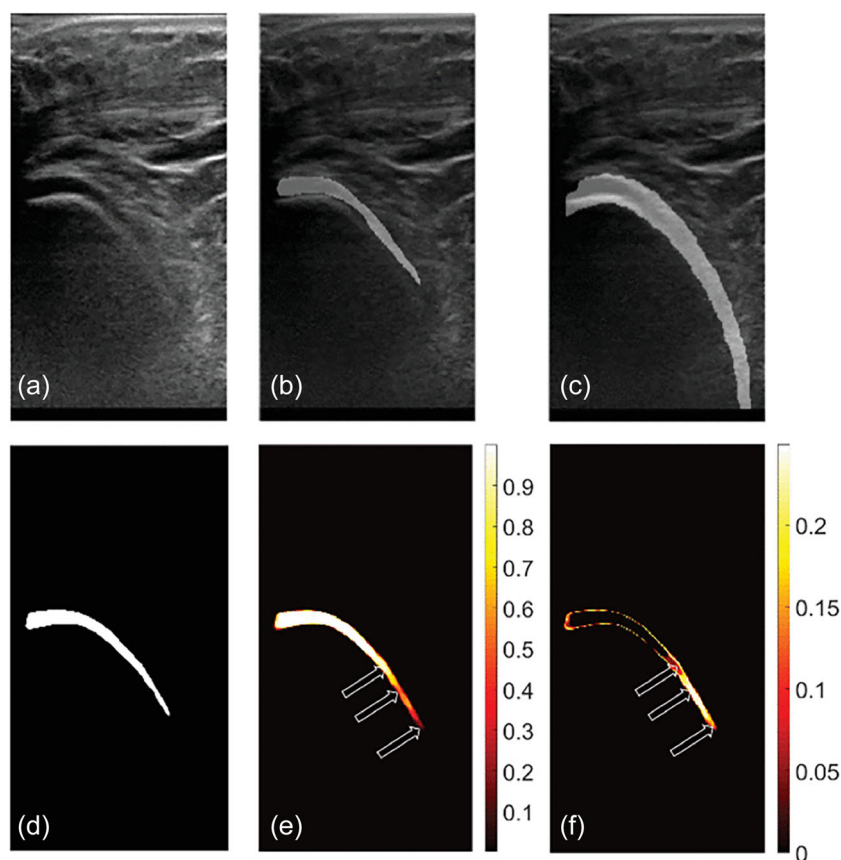


FIGURE 14 Figure from reference [6], showcasing a sample image, its label, predictions and uncertainty heatmap. (a) Input ultrasound image. (b) Label acquired from ultrasound image. (c) MRI-based label. (d) Baseline U-Net prediction. (e) U-Net with MC dropout predictions. (f) Uncertainty heatmap based on the variance of the predictions in (f). Notice how the arrows in (d) and (e) correspond to the difference between (b) and (c).

Skeletal muscle segmentation in hip CT images

Hiasa et al. [40] investigated the use of MC dropout in a U-Net model developed to segment skeletal muscles CT images of the hip and its benefits in estimating predictive confidence (Figure 15). If accurate and reliable, such a model is highly useful in various applications, such as biomechanical simulations and quantitative muscle atrophy evaluation.

They found that uncertainty measures inversely correlated with segmentation accuracy for most muscles. This correlation indicates that UQ can help identify cases where the model may be less reliable, prompting clinicians to exercise caution or seek additional input.

Moreover, UQ can play a significant role in optimizing the annotation process through active learning. Annotating medical images is time-consuming and requires expert knowledge. By using uncertainty measures to identify the most informative or uncertain cases, as proposed by Hiasa et al., researchers can prioritize which images need expert annotation. This approach can accelerate the development of AI models by efficiently utilizing limited expert man-hours.

Total hip arthroplasty (THA) automatic implant detection

Rouzrokh et al. [92] designed an automated framework to detect and classify the design of femoral and acetabular components of THA implants in hip radiographs. Such a framework improves revision THA surgery planning, which necessitates the preoperative identification of in situ implants using radiographs.

Their framework included a DL model that separately classifies the design of the femoral and acetabular components, which was first calibrated (Section 'Direct predictive confidence' [72]) and then complemented with a two-component UQ method combining CP (Section 'Conformal prediction', Figure 16) and a HLA method (Section 'Hidden layer analysis') used to detect test subjects that might be outside the distribution which the model is trained on, using both the hidden layers and non-conformity scores. This secondary OOD detection method was implemented to address the shortcomings of UQ methods in estimating predictive confidence on test data, which is significantly different from the training data [59, 81].

In addition, they also used an XAI method called integrated gradients [67] which, as explained earlier,

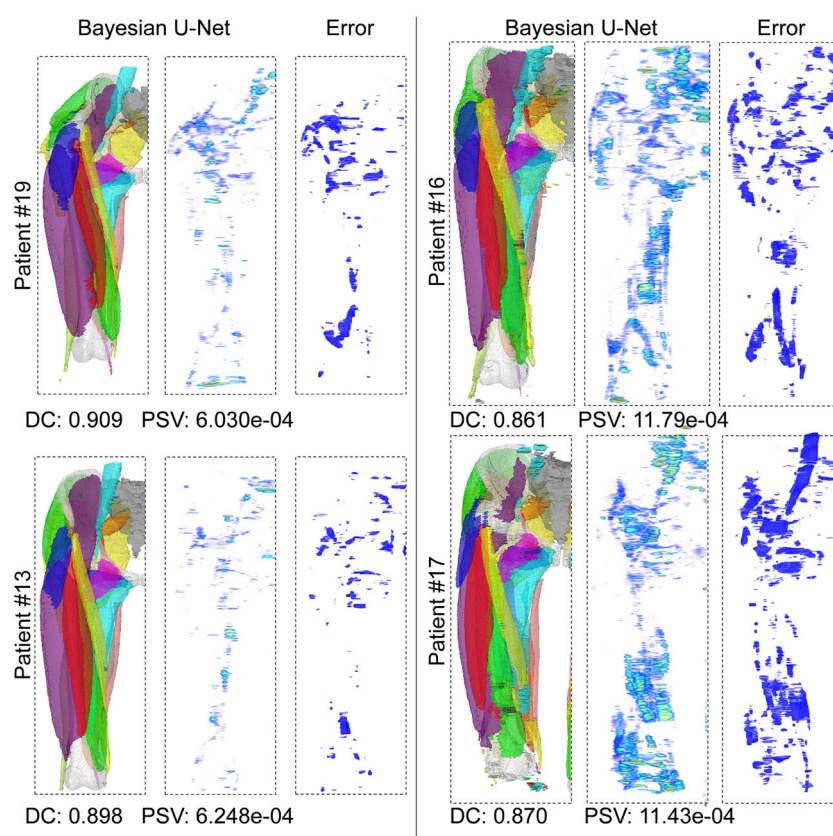


FIGURE 15 Diagram from reference [40], showcasing predictions for four subjects. For each subject, the columns from left to right are prediction, uncertainty heatmaps and errors in the prediction compared to the label. DC, Dice score; PSV, predicted structural variance, which estimates uncertainty.



FIGURE 16 Sample prediction sets from reference [92] (license number: 5855210427073). (a) Most certain prediction sets, with single-member prediction sets indicative of high confidence in the predictions. (b) Most uncertain prediction sets, which have multiple members in each set, indicative of high uncertainty. Notice how these images are cropped, use suboptimal postures, or do not have the entirety of the implant captured in the image.

highlights the areas of the image that impact the model's prediction the most, in order to communicate the model's decision-making process to the clinician user.

Their inference pipeline consists of the following (Figure 17):

1. The input image is first classified as definitive outlier, probable outlier or a definitive inlier (named 'traffic light framework').
2. In accordance with the CP methodology, the model outputs two prediction sets, one for the femoral (stem) and one for the acetabular (cup) component which contains the ground-truth class with an arbitrary probability chosen during calibration (Section 'Conformal prediction').
3. The framework also outputs visual XAI outputs, as described earlier.

In their evaluations, the authors reported high coverage for their CP method in both internal and external

validation (97%–99.8%), although efficiency deteriorated in external validation (only 69.8% of the prediction sets consisted of a single predicted class, compared to 84.3%–94.8% in internal validation).

This study successfully highlights the potentials of integrated approaches incorporating UQ, OOD detection, and XAI along with large medical data sets, which in the case of this study consisted of 241,419 radiographs, to enhance the reliability and explainability of predictive confidence/uncertainty estimation in medical DL models.

DISCUSSION

Despite these advancements, challenges remain in applying UQ effectively in MSK medicine. Computational constraints are a significant consideration, as some UQ methods increase the computational load, which may not be practical in real-time clinical environments. Data requirements pose another

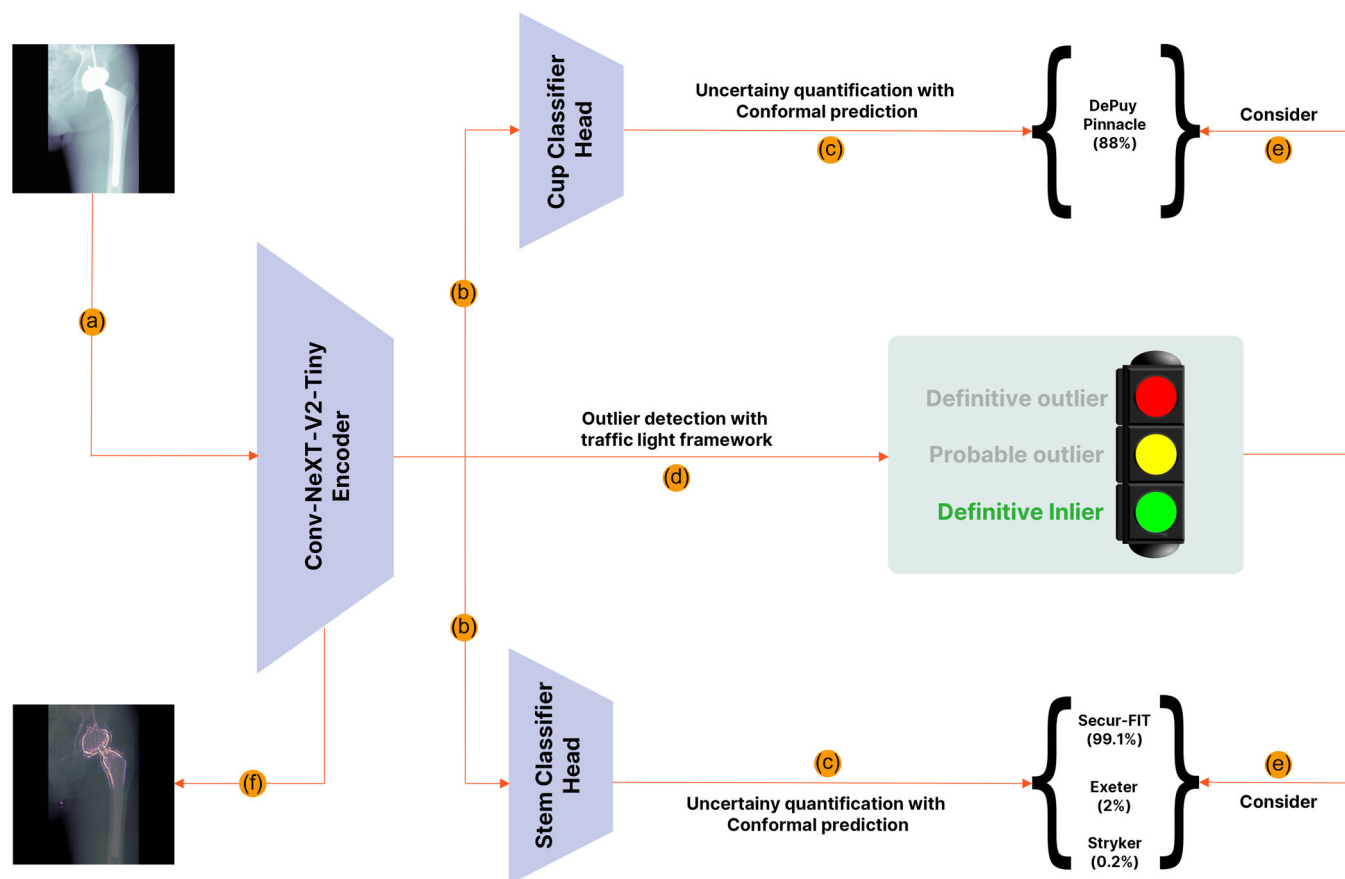


FIGURE 17 Framework diagram from reference [92] (license number: 5855210427073). (a) Input image. (b) Femoral (stem) and acetabular (cup) classifiers. (c) Femoral (stem) and acetabular (cup) prediction sets. (d) 'Traffic light' framework for OOD detection, which communicates overall levels of confidence. (f) Integrated gradients overlay heatmap, which enhances explainability.

challenge, especially when methods necessitate additional calibration data sets or multiple expert annotations. Developing standardized evaluation metrics for UQ is also essential to assess the effectiveness of different approaches and understand their clinical impact.

Addressing these challenges requires a collaborative effort between AI researchers and clinicians. User-centred design principles should guide the development of AI models, ensuring that uncertainty estimates are presented in a way that is intuitive and actionable for clinicians. Educating clinicians about the interpretation of uncertainty measures is equally important, as misinterpretation could lead to inappropriate clinical decisions.

Looking forward, research should focus on developing robust UQ methods that can handle OOD data effectively. The clinical environment is dynamic, with variations in patient populations and imaging practices. AI models must be adaptable and capable of recognizing when they encounter unfamiliar cases. Additionally, integrating UQ into clinical workflows requires consideration of usability and efficiency. Methods that add minimal computational overhead

and fit seamlessly into existing practices are more likely to be adopted.

In a later work, we also intend to discuss how UQ methods and trustworthy AI frameworks in general are actually supposed to function in their intended usage scenario alongside human experts. Since the end goal for these DL frameworks is integration in clinical decision-making, several aspects, such as instances involving expert versus model disagreement (with or without high uncertainty) and how to solve these disagreements, warrant further investigation.

CONCLUSION

To summarise, UQ is not just a technical enhancement but a clinical imperative for AI models in MSK medicine. By providing reliable estimates of predictive confidence, UQ aligns AI models with the nuanced decision-making processes of clinicians. While challenges exist, the potential benefits in improving patient care and advancing the integration of AI into clinical practice are substantial. Continued collaboration and focused research in this area are essential steps toward

realizing the full potential of AI in enhancing MSK healthcare.

AUTHOR CONTRIBUTIONS

All authors contributed to the study conception and design. Amir M. Vahdani and Moein Shariatnia contributed equally to writing the first draft of the manuscript. All authors reviewed the manuscript and provided feedback for further revisions. Amir M. Vahdani prepared the visualizations. All authors reviewed the finalized manuscript and approved of publication.

ACKNOWLEDGEMENTS

The authors would like to thank Dr. Carlos Fernandez-Granda for his comments on the technical aspects of this work.

CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

Data sharing is not applicable to this article as no data sets were generated or analyzed during the current study.

ETHICS STATEMENT

This study reviews peer-reviewed publications. For ethics approvals related to each original study, please refer to their related papers. Please refer to the cited papers for any relevant material on consent to participate for participants.

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How to cite this article: Vahdani AM, Shariatnia M, Rajpurkar P, Pareek A. Towards trustworthy artificial intelligence in musculoskeletal medicine: A narrative review on uncertainty quantification. *Knee Surg Sports Traumatol Arthrosc*. 2025; 33:3418–37. <https://doi.org/10.1002/ksa.12737>

APPENDIX A: DEFINITIONS

- Deep learning (DL) model: A type of machine learning model that consists of compositions of mathematical operations, generally stacked in a sequential fashion (*layers*), including an *input layer*, several *hidden layers* and an *output layer*. A common example of these mathematical operations is $x: wx + b$; here, w and b are examples of *parameters*, which the model learns during its *training*. The exact arrangement, shape and depth of a DL model is known as its *architecture*. These models were initially developed with the aim of mimicking biological nervous systems, hence their older name, 'neural network'. The scope of discussion in this article is focused on *supervised DL*.
- Training: The main stage of developing DL models, which consists of iteratively learning from a special subset of the available data called the training data set. The goal here is to generalize to previously unseen data, known as test data. Readers proficient with statistical modelling procedures common in medical research are probably familiar with the terms

- 'train', 'internal validation' and 'external validation' data. With a few caveats, these terms could be considered as roughly equivalent to 'train', 'validation' and 'test' data from DL literature.
- Inference: The actual usage phase of a DL model, in which the model is utilized to output individual predictions for previously unseen subjects. In technical DL literature, inference can have a broader meaning based on thermodynamics perspectives [60, 87], which is beyond the scope of this paper.
 - Post hoc: We use this term to refer to any further changes to a DL model after its training stage, but before the inference stage.
 - Classification task: Categorizing input subjects into predefined classes; assigning CT scan images of joint transplants to either loose or non-loose is an example of a classification task with two classes ($k=2$). Prediction performance in these tasks is evaluated using metrics such as accuracy, sensitivity, specificity and AUC-ROC.
 - Segmentation task: Classifying different areas or regions of interest in an image into distinct regions based on pre-specified classes; for example, delineating a hand radiograph into background, carpal, metacarpal and phalangeal regions. Prediction performance in these tasks can be measured using classification metrics, but conventionally specific segmentation metrics such as the Dice metric (used for assessing area/volume overlap) and contour metrics (assessing how well the contours of the ground-truth and the prediction align) are used to evaluate segmentation models.
 - Dice score: A metric that measures the overlap between two areas in an image, used in evaluating segmentation models when comparing the predicted area vs. the ground truth area.
 - Dropout: A special technique used in training DL models in which random subsets of the units in specific layers are deactivated, helping the model adapt to noise and heterogeneity. Conventionally, this technique is only used during training, and not during inference.
 - Ground-truth: The objective reality that we aim to capture or predict with our models or tests, also called *label* when dealing with data sets used to train or test DL models; closely related to 'gold standard' in medical nomenclature, which refers to the most reliable measurement/diagnosis for a subject.
 - U-net: A type of DL model architecture which is widely used in segmentation tasks, first developed by reference [89].
 - Active learning: Active learning is a machine learning approach that efficiently builds accurate models while minimizing the need for extensive manual data labelling. In this method, the most informative or uncertain cases from a large set of unlabelled data are identified, and expert manual input is only required for these high-priority cases. In the context of DL models on medical imaging data, this equates to learning effectively from a smaller number of carefully chosen examples, reducing time and resource requirements in developing AI systems for healthcare applications.